

Special Session Proposal - E2A 2027

Title of the Proposal: Intelligence-Driven Signal and Image Processing for Secure, Intelligent, and Trustworthy VLSI Systems

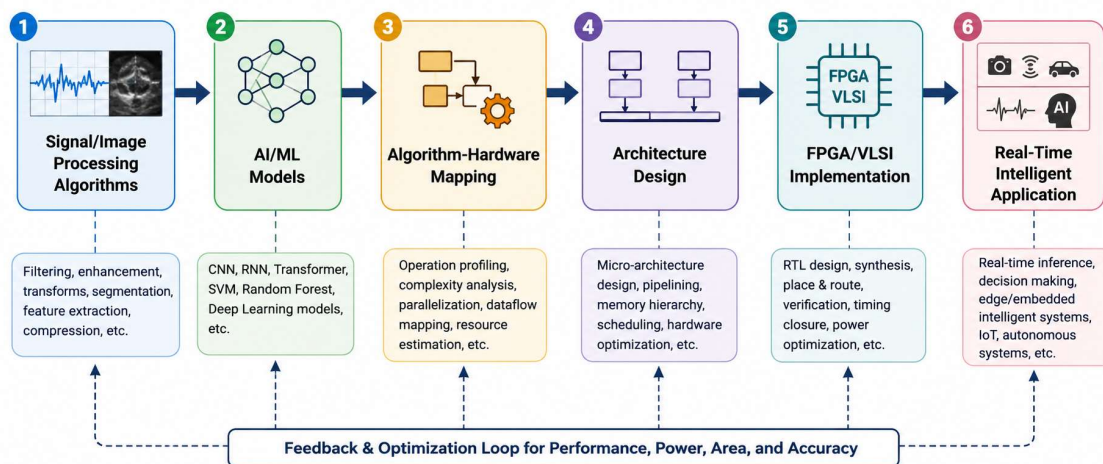
A Brief Description of the Proposed Area:

The rapid convergence of Artificial Intelligence (AI), signal and image processing, and Very-Large-Scale Integration (VLSI) technology is transforming the design of intelligent electronic systems for next-generation applications. The increasing demand for real-time vision, secure data processing, intelligent sensing, edge computing, and autonomous decision-making has created a strong need for hardware architectures capable of executing computationally intensive AI and signal/image processing algorithms with low latency, low power consumption, compact hardware resources, and enhanced security.

Modern applications such as autonomous vehicles, intelligent transportation systems, medical imaging, smart surveillance, UAVs, robotics, industrial automation, IoT, and edge intelligence require the integration of sophisticated signal and image processing algorithms with efficient hardware platforms. Conventional software-only implementations often face limitations in terms of computational latency, energy consumption, memory bandwidth, and real-time processing capability. Consequently, FPGA- and VLSI-based implementations, AI accelerators, hardware-software co-design, and application-specific architectures have emerged as important approaches for realizing intelligent systems at the edge.

The session aims to provide a focused platform for researchers, academicians, and industry professionals working at the intersection of signal processing, image processing, artificial intelligence, machine learning, deep learning, computer vision, multimodal and deepfake detection, VLSI design, FPGA implementation, embedded systems, and hardware security. Particular emphasis is placed on trustworthy multimedia intelligence, including the detection, localization, and analysis of AI-generated and manipulated images, videos, audio, and multimodal content.

The session particularly emphasizes the complete development pipeline:



Rationale for Organizing this Special Session:

AI-based signal and image processing has become a fundamental component of modern intelligent systems. At the same time, the rapid growth of generative AI has made highly realistic deepfakes and synthetic multimedia increasingly difficult to distinguish from authentic content. Deepfake detection therefore requires robust analysis of spatial, temporal, frequency-domain, physiological, semantic, and multimodal inconsistencies while maintaining low computational complexity and real-time performance. The increasing complexity of deep neural networks, high-resolution multimedia data, and real-time sensing applications creates significant challenges for conventional computing platforms. Computational complexity, memory requirements, power consumption, processing latency, and data movement are particularly critical for edge and embedded applications.

VLSI and FPGA technologies provide an effective pathway for overcoming these limitations by enabling dedicated, parallel, and application-specific hardware architectures. Hardware implementation of AI and signal/image processing algorithms can provide significant improvements in throughput, latency, energy efficiency, and real-time performance. At the same time, the growing deployment of intelligent systems in security-sensitive applications has created an urgent need for secure hardware architectures capable of protecting data, models, communication, and computation.

Existing research is frequently distributed across separate communities, including image processing, machine learning, VLSI design, FPGA systems, hardware security, embedded computing, and AI accelerators. As a result, there is a need for a dedicated forum that connects these complementary areas and promotes the complete transition from intelligent algorithms to practical hardware implementation.

This special session is proposed to address this research gap by bringing together researchers working on AI-enabled signal and image processing, Deepfake detection and multimedia forensics, multimodal learning, hardware-efficient machine learning, VLSI architectures, FPGA acceleration, secure multimedia processing, and edge intelligence. The session will particularly encourage research that demonstrates a clear relationship between algorithmic innovation and hardware implementation, including hardware-aware Deepfake detection models, efficient multimodal architectures, explainable detection systems, and real-time forensic pipelines.

The session will therefore provide a unique interdisciplinary platform for discussing the challenges and opportunities associated with designing secure, intelligent, hardware-efficient, and application-oriented VLSI systems.

Technical Outline of the Session and Topics

- Deepfake detection, localization, and multimedia forensic algorithms
- Multimodal and explainable AI for trustworthy synthetic-media detection
- Hardware-efficient deepfake detection and forensic models
- Novel AI-enabled signal/image processing algorithms
- Hardware-efficient AI/ML models
- Low-power and energy-efficient architectures

- Real-time processing capabilities
- Hardware Implementation of Signal Processing Algorithms
- Secure and trusted hardware architectures
- AI accelerators for edge applications
- Low-Power and Energy-Efficient Image Processing
- Prototype and hardware validation
- Approximate Computing for Multimedia and AI
- Application-oriented intelligent embedded systems

The session will foster interdisciplinary collaboration between researchers in VLSI, electronics, signal processing, image processing, artificial intelligence, computer vision, embedded systems, cybersecurity, and hardware design, thereby promoting the development of next-generation secure and intelligent computing platforms.

Details of Session Organizers

Organizer 1



Name: Dr. M. Suresh

Affiliation: School of Electrical and Electronics Engineering, VIT Bhopal University

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Brief Bio: Dr. M Suresh is currently working as an Associate Professor in the School of Electrical and Electronics Engineering (SEEE) at VIT Bhopal University. I completed my PhD in Digital Design from SUIIT, India in 2021. I holds a Master's in Engineering (ME) from Jadavpur University and Bachelor's in Engineering (BE) from Andhra University in ECE. With over 25 years in academic and research, he has made significant contributions. I have a strong publication record with numerous papers published in repute journals and conferences. He also played pivotal role in developing employment-oriented courses, such as VLSI design engineering and FPGA design. In addition, he has mentored several undergraduate and post graduate projects nurturing the next generation of engineers and researchers. I am an active member of professional organizations, including being life member of Indian Society for Technical Education (ISTE), the Institution of Electronics and Telecommunication Engineers (IETE), and a member of Institute of Electrical and Electronics Engineers (IEEE). His primary Research interests encompass Digital design, FPGA design, and Microprocessor based system design reflecting his dedication to advancing the field and sharing his expertise with others. I

also actively contribute to the organization of international conferences and serve as General Chair and session chair of ICSDP-2024-25 and IECS-2026.

Organizer 2



Name: Dr. Vijay Birichha

Affiliation: SCAI, VIT Bhopal University

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Brief Bio: : Dr. Vijay Birichha is currently working as an Assistant Professor in the at SCAI VIT Bhopal University. I received a Ph.D. in Computer Engineering from Devi Ahilya Vishwavidyalaya (DAVV), Indore, India. I earned his Master of Engineering (M.E.) in Computer Engineering with a specialization in Software Engineering from the Institute of Engineering and Technology (IET), DAVV, Indore, and a Bachelor of Engineering (B.E.) in Computer Science and Engineering from Guru Ghasidas Vishwavidyalaya, Bilaspur. With over 18 years of teaching experience in computer engineering, I specializes in applying Artificial Intelligence to healthcare, particularly in breast cancer detection and classification. His research interests also include computer vision, machine learning, and software engineering. I have published more than 20 research papers in peer-reviewed journals and holds over five patents and design publications. Additionally, I had completed professional certifications in IBM DB2, Tivoli, and Rational Unified Process (RUP). In 2008, he received the IBM "The Great Mind Challenge" Mentor Award, titled "GURU," for his contributions.

Organizer 3



Name: Dr. Amit Kumar Singh

Affiliation: School of Electrical and Electronics Engineering, VIT Bhopal University

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Brief Bio: : Dr. Amit Kumar Singh is currently working as an Assistant Professor in the School of Electrical and Electronics Engineering (SEEE) at VIT Bhopal University. I received B.Tech. in Electronics and Communication Engineering from FGIET Raebareli, UP, India, in 2009.

M.Tech. in Digital Techniques and Instrumentation from Electronics Engineering Department of IIT- Varanasi (BHU), Varanasi, India, in 2015. I had done Ph.D. from the department of ECE, MNNIT Allahabad, Prayagraj India. I have about ten years of teaching and research experience. He has published nearly 7 papers in peer-reviewed international journals and conferences. His current research interest includes genomic signal processing with a special focus on DSP techniques for the identification of protein-coding regions in DNA sequences, Image Processing, Data Hiding, and Bioinformatics. I also actively contribute to the organization of international conferences and serve as Co-convener and session chair of ICSDP-2025.

Organizer 4



Name: Dr. Anurag Tiwari

Affiliation: VIT Bhopal University, Bhopal, Madhya Pradesh

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Brief Bio: Dr. Anurag Tiwari is currently working as an Assistant Professor in the School of Electrical and Electronics Engineering (SEEE) at VIT Bhopal University. I received my B.Tech. degree in Electronics and Communication Engineering from Rewa Engineering College, Rewa, India, and my M.Tech degree in Digital Systems from Motilal Nehru National Institute of Technology (MNNIT), Allahabad, India. I completed my Ph.D. at MNNIT Allahabad, India, with a specialization in Digital Image Watermarking.

I am actively engaged in research focusing on digital image watermarking, data security, and multimedia security. My research interests include Signal and Image Processing, Digital Image Watermarking, Data Hiding, Cryptography, Machine Learning, and Deep Learning. I have authored and co-authored more than 20 research publications in reputed international journals, conferences, and book chapters. I also actively contribute to the organization of international conferences and serve as Co-convener and session chair of ICSDP-2025.

- **Potential Reviewer Profile**

Potential reviewers may be invited from diverse but complementary research areas, including:

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